

Business Rules Applied

Complete catalog of the validation rules enforced by the Master Data Validator. Each rule specifies what it checks, what fails it, and the suggested remediation.

This document maps each rule to its SAP field and shows concrete examples of values that fail vs. pass. Use this reference when deciding how to handle specific decisions in the tool, and when configuring rule-specific thresholds for other LTMC objects.

Severity Levels

Error — Must be fixed before LTMC upload. The record will fail LTMC validation or produce incorrect master data. **Warning** — Should be reviewed but may pass LTMC. Typically indicates data that is technically valid but semantically suspect.

Rule Catalog

1. Mandatory Field Missing

ERROR

Rule ID: mandatory_missing SAP field: KNA1-KUNNR, KNA1-NAME1, KNA1-LAND1, etc.

What it checks: Fields required by SAP to create a valid master record are blank.

Fails (example)	Passes (example)
Customer Number = "" Name = "" Country = ""	Customer Number = "6000123" Name = "Siya Medical" Country = "IN"

Recommended action: Fill all affected rows with the correct value. Use the bulk *Fill with* action when the correct value is the same across all rows, otherwise edit individually.

2. Length Exceeded

ERROR

Rule ID: length_exceeded SAP field: Varies by field – defined in SAP data dictionary

What it checks: A field contains more characters than the SAP field width allows. Caused by descriptive values pasted into code fields, or free-text put into short identifier fields.

Fails (example)	Passes (example)
Inco. Location1 = "Cost Insurance and Freight" (26 chars, max 10)	Inco. Location1 = "CIF" (3 chars)

Recommended action: Use *Replace with* when a single standard code replaces all variants (e.g., all rows to "CIF"). Use *Truncate all* to hard-cut to the max length when trailing detail is acceptable to lose.

3. Duplicate Record

ERROR

Rule ID: duplicate_record SAP field: KNA1-KUNNR (Customer), MARA-MATNR (Material)

What it checks: Two or more rows share the same primary key. In customer data this is Customer Number; in material data this is Material Number. Duplicates are rejected outright by LTMC.

Fails (example)

Row 12: Customer Number = "6000123" .
Name = "Siya Medical"
Row 47: Customer Number = "6000123" .
Name = "Siya Med"

Passes (example)

Only one row per unique Customer Number

Recommended action: Use *Preview affected rows* to see all duplicates side-by-side with key fields (Name, City, PAN, GSTIN) and their Sales Areas (SO/DC/DIV). Select rows with checkboxes and use *Delete selected*, or use *Delete duplicate rows* with a keep strategy (keep first / keep last / keep most complete).

4. Incoterms Location Contains Description

ERROR

Rule ID: inco_location_description SAP field: KNVV-INC01

What it checks: Incoterms Location field contains the full incoterm description rather than the 3-letter code. Typical of data copied from purchase orders.

Fails (example)

Inco. Location1 = "Cost Insurance and Freight"
Inco. Location1 = "Free On Board – Mumbai Port"

Passes (example)

Inco. Location1 = "CIF"
Inco. Location1 = "FOB"

Recommended action: Apply *Replace with* and enter the standard 3-letter code. All rows with the same description collapse into one decision.

5. Junk / Placeholder Value

WARNING

Rule ID: junk_value SAP field: Varies – any text field

What it checks: Field contains obvious placeholder text that shouldn't survive to production: "TBD", "XXX", "TEMP", "TEST", "DUMMY", etc.

Fails (example)

Name 2 = "TBD"
City = "XXX"
PAN Number = "TEMP"

Passes (example)

Name 2 = "" (blank is preferable to junk)
City = "Mumbai"
PAN Number = "AAAPL1234C"

Recommended action: Use *Clear all* to blank them out (if the field is optional), or *Replace with* if you have the correct values. Document the choice in the business reason.

6. GSTIN Format Invalid

ERROR

Rule ID: gstin_format SAP field: J_1IMOCUST-J_1ILSTNO or custom extension

What it checks: Indian Goods and Services Tax Identification Number doesn't match the 15-character format: 2 state digits + 10-char PAN + 1 entity digit + 1 "Z" + 1 checksum.

Fails (example)	Passes (example)
GSTIN = "24AAMFN9348L1Z" (14 chars) GSTIN = "ABCDE1234F1Z5" (invalid state digits)	GSTIN = "27AAMFN9348L1Z5"

Recommended action: For unregistered vendors/customers, use *Set all to URP* (Unregistered Person). Otherwise review individually — this usually means the source data is wrong and needs to be fixed at the source.

7. PAN Format Invalid

ERROR

Rule ID: pan_format SAP field: J_1IPANREF-J_1IPANNO or extension

What it checks: Indian Permanent Account Number doesn't match the 10-character format: 5 letters + 4 digits + 1 letter (e.g., AAAPL1234C).

Fails (example)	Passes (example)
PAN = "ABCDE1234" (9 chars) PAN = "12345ABCDE" (wrong order)	PAN = "AAAPL1234C"

Recommended action: Set to *URP* for unregistered entities, otherwise correct individually at source.

8. GSTIN Length

ERROR

Rule ID: gstin_length SAP field: J_1IMOCUST-J_1ILSTNO

What it checks: GSTIN isn't 15 characters (separate check from format — catches truncation vs. malformed data).

Fails (example)	Passes (example)
GSTIN = "27AAMFN9348L1Z" (14 chars)	GSTIN = "27AAMFN9348L1Z5" (15 chars)

Recommended action: Identical remediation to *gstin_format*: set URP for unregistered, or correct at source.

9. Drug License Expired

WARNING

Rule ID: dl_expired SAP field: Custom extension on KNA1 or KNVV

What it checks: Drug License date is in the past. Relevant for medical device and pharma customers — expired licenses must be renewed before shipment.

Fails (example)	Passes (example)
DL Expiry = "01-Jan-2024" (past)	DL Expiry = "31-Dec-2026" (future, valid)

Recommended action: Review each — these cannot be bulk-fixed because each customer's DL renewal is individual. Use *Preview affected rows* and either update the expiry date from renewed documentation or flag the customer as inactive.

10. Invalid State Code (India)

ERROR

Rule ID: invalid_state_in SAP field: KNA1-REGIO

What it checks: Indian state code doesn't match a valid 2-digit state code as defined in T005S (SAP region table).

Fails (example)	Passes (example)
State = "99" (not a valid Indian state code) State = "XX"	State = "27" (Maharashtra), "29" (Karnataka), "33" (Tamil Nadu), etc.

Recommended action: Review each. State codes are usually typos or postal code confusion. Correct to the matching 2-digit code from T005S. Cannot be bulk-fixed because each row has a different correct answer.

Rule Configuration

Rules are defined in code today (*backend/services/validator.py*). For production, rules will be driven by a configuration table in Postgres so new checks can be added without a deployment. The rule catalog is not final — additional rules will be added as new LTMC objects (Material, BOM, Vendor) are brought into the tool.

Pattern vs. Individual Decisions

The tool groups errors into decisions using a threshold: when 5 or more rows share the same rule + column + value, they become a **Pattern** decision with a bulk action. Fewer than 5 matching rows become an **Individual** decision requiring case-by-case review. The threshold is configurable in *decision_engine.py*.

Audit of Rule Decisions

Every bulk action requires a business reason. This reason is stored in the audit trail alongside the user, timestamp, affected row count, and rule. During SAP validation or regulatory review, the audit trail can be exported by the administrator.